

Methods Guide

HD-sEMG Volume-Conductor Forward Model: a reproducible protocol for building an imaging-derived 3D forearm tissue model and a first-principles surface-EMG forward simulation.

Part A: Anatomical Model (3D Slicer)

A1. Acquire forearm imaging

Source: Visible Human Project male dataset, PNG cryosection format. The axial cryosections are continuously numbered head→foot; the arms lie alongside the torso, so the mid-forearm cross-section (radius + ulna inside a full muscle ring) appears in the lower-abdomen slice range.

Block used: files a_vm1650–a_vm1710 (folder abdomen/), ~61 axial slices spanning ~6 cm of forearm. Scale: 0.33 mm in-plane (X/Y), 1.0 mm between slices (Z).

A2. Reconstruct the 3D volume

3D Slicer + SlicerMorph → ImageStacks module. Select the first slice as archetype; load at full resolution, grayscale. Set image spacing to $0.33 \times 0.33 \times 1.0$ mm so the geometry is anatomically accurate.

A3. Isolate a single forearm

Crop Volume module → create an ROI box → wrap it around one forearm cross-section (excluding the torso) → Apply.

A4. Segment the four tissues (Segment Editor)

Tissue	Method
Muscle	Threshold ≈ 14 –99 (mid-grey interior); Apply.
Bone	Threshold ≈ 180 –244 (brightest); Apply; then Islands → Keep selected island, click the bone.
Fat	Threshold ≈ 180 –244; masking = Allow overlap; Apply (keeps subcutaneous + intermuscular fat).
Skin	Threshold ≈ 14 –244 (whole limb); Islands → Keep largest island; then Hollow effect, 1.0 mm, 'outside surface' → thin outer shell.

Note: bone and fat share a brightness range, so they are separated by spatial connectivity (Islands), not by threshold alone.

A5. Export meshes

Segmentations module → Representations → Closed surface → Update (select the conversion path before converting). Then Export to files → STL, one file per tissue.

Part B: Visualization (Blender)

B1. Import & optimize

Import all four STL tissues. The muscle mesh is very high-poly; apply a Decimate modifier (ratio ≈ 0.15). Shade Smooth all meshes.

B2. Materials (Principled BSDF)

Tissue	Base color (hex)	Notes
Muscle	8C2F2F	Subsurface weight ≈ 0.4 , reddish subsurface radius
Bone	EDE6D3	Matte (roughness ≈ 0.7)
Fat	E9D08A	Waxy, slight subsurface
Skin	D9A684	Subsurface; optionally translucent (lower alpha + Blended)

Subsurface scattering is essential; it is what makes tissue read as wet/organic rather than as plastic.

B3. Camera, lighting, render

3/4-angle camera aimed at the cross-section, framed on the model bounds. Three-point lighting using sun lamps (direction-based, so independent of scene scale). Render engine: Cycles, with denoising enabled. Optional polish: depth of field (focus on the cross-section), F-stop ≈ 4 .

Part C: Physics Simulation (Python)

File: sfap_model.py (requires numpy, matplotlib).

C1. Source: Rosenfalck action potential & current source

Intracellular action potential along the fiber:

$$V_m(z) = A \cdot z^3 \cdot e^{-\lambda z} - B \text{ for } z \geq 0 \quad (A = 96 \text{ mV/mm}^3, \lambda = 1 / \text{mm}, B = 90 \text{ mV}).$$

The transmembrane current source that radiates is proportional to the second spatial derivative of V_m , a triphasic (tripole) profile.

C2. Volume conductor: detected single-fiber action potential

A point current source produces potential $\phi = I / (4\pi\sigma r)$, where σ is tissue conductivity and r the source-to-electrode distance. The fiber is a line of such sources; sum their contributions to one surface electrode while letting the action potential travel (front position $p = v \cdot t$, conduction velocity $v \approx 4 \text{ m/s}$, fiber depth h). Result: a triphasic detected SFAP waveform vs. time, a simulated sEMG signal.

C3. Electrode grid: spatial map, crosstalk, depth blurring

Replace the single electrode with a 2D grid on the skin and place two or more motor units at different lateral positions and depths. Compute the surface potential map (snapshot) per motor unit. Two phenomena emerge directly from the physics:

Phenomenon	What it shows
Depth / fat blurring	A deeper source produces a weaker, more spread-out surface map.
Crosstalk	One muscle's field appears on electrodes over a neighboring muscle; quantified as the ratio of crosstalk amplitude to wanted-signal amplitude at an electrode.

Scientific Basis

- Rosenfalck, P. (1969). Intra- and extracellular potential fields of active nerve and muscle fibres. *Acta Physiologica Scandinavica*, Suppl. 321.
- Farina, D. & Merletti, R. (2001). A novel approach for precise simulation of the EMG signal detected by surface electrodes. *IEEE Trans. Biomed. Eng.* 48(6).

- Merletti, R. & Farina, D. (2016). *Surface Electromyography: Physiology, Engineering, and Applications*. Wiley-IEEE Press.
- Imaging: U.S. National Library of Medicine, Visible Human Project (public domain).